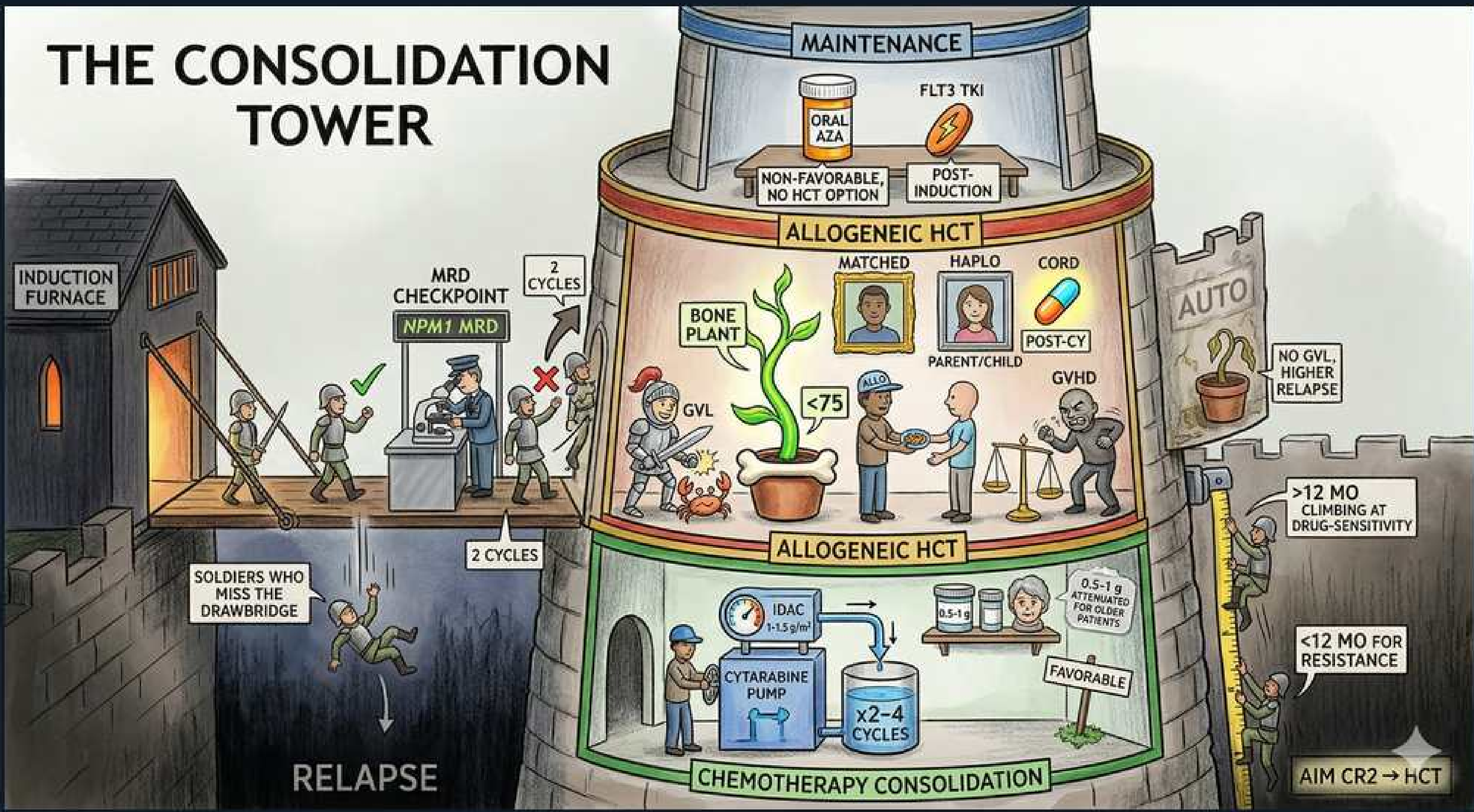
AML — Consolidation & Transplant



1. Consolidation tower

WHAT YOU SEE

A tall stone fortress titled THE CONSOLIDATION TOWER — the multi-tiered tower is post-remission therapy built on top of induction to prevent relapse.

WHAT IT ENCODES

After induction achieves complete remission (CR), post-remission (consolidation) therapy is mandatory to eradicate measurable residual disease and prevent relapse — CR alone is not cure. The choice of consolidation is driven by ELN risk: favorable → HiDAC chemotherapy; intermediate/adverse → allogeneic HCT in CR1 if eligible.

BOARD TRAP

Achieving CR after induction is NOT cure; without consolidation virtually all patients relapse. The consolidation modality is dictated by genetic risk group, not by how easily CR was attained.

2. Induction furnace

WHAT YOU SEE

A glowing furnace at the tower's base feeding it (INDUCTION FURNACE) — the fire below is 7+3 induction that must achieve CR before consolidation begins.

WHAT IT ENCODES

Consolidation begins only after induction (classically 7+3: 7 days continuous cytarabine + 3 days of an anthracycline, daunorubicin or idarubicin) achieves CR. CR requires <5% marrow blasts, neutrophil recovery >1,000, and platelets >100,000. Fit patients get intensive 7+3 (± midostaurin if FLT3+, ± gemtuzumab if CD33+/favorable); unfit/older patients get azacitidine + venetoclax.

BOARD TRAP

7+3 = 7 days cytarabine + 3 days anthracycline; do not reverse the numbers. Older/unfit patients are not abandoned — azacitidine + venetoclax is the standard non-intensive induction.

3. MRD checkpoint

WHAT YOU SEE

A guard post with a gate reading NPM1 MRD where soldiers are screened — the checkpoint is residual-disease testing that can reroute a 'favorable' patient to transplant.

WHAT IT ENCODES

Measurable residual disease (by multiparameter flow cytometry or molecular PCR, e.g. NPM1 or CBF transcripts) is assessed after induction/consolidation. MRD status refines risk: MRD-positive patients have high relapse rates and are pushed toward allo-HCT even if their genetics were favorable.

BOARD TRAP

MRD-positivity in morphologic CR still carries a high relapse risk — it can reclassify a 'favorable' patient toward transplant and is not negated by a normal-appearing marrow.

4. Relapse / missed drawbridge

WHAT YOU SEE

Soldiers tumbling off a broken drawbridge into the abyss labeled RELAPSE — the men who 'miss the bridge' are patients who relapse when consolidation is skipped.

WHAT IT ENCODES

Inadequate or omitted consolidation leads to relapse. Relapse is defined as ≥5% marrow blasts, reappearance of blasts in blood, or new extramedullary disease after CR. Relapsed AML has a poor prognosis and requires reinduction with intent to reach CR2 and proceed to transplant.

5. HiDAC consolidation

WHAT YOU SEE

A cytarabine pump in the green chemo zone marked x2-4 CYCLES — the pump cycling Ara-C is HiDAC consolidation for favorable-risk AML.

WHAT IT ENCODES

Favorable-risk AML is consolidated with high-dose cytarabine (HiDAC) for ~2-4 cycles, generally without transplant in CR1. Watch HiDAC toxicities: cerebellar ataxia/neurotoxicity (do a cerebellar exam before each dose, especially in renal impairment/elderly), conjunctivitis (give prophylactic steroid eye drops), and palmar-plantar erythrodysesthesia.

BOARD TRAP

Favorable-risk AML is consolidated with HiDAC and generally NOT transplanted in CR1. Cerebellar toxicity from HiDAC mandates a neuro exam before dosing — hold the drug if ataxia/nystagmus appears.

6. Allogeneic HCT

WHAT YOU SEE

An ALLOGENEIC HCT band around the tower with a bone-marrow plant being potted (BONE PLANT) — the transplanted 'plant' is donor allo-HCT for intermediate/adverse risk.

WHAT IT ENCODES

Allogeneic (donor) HCT in CR1 is the consolidation of choice for intermediate- and adverse-risk AML in eligible patients, because the graft-versus-leukemia effect provides curative potential beyond chemotherapy. AUTOLOGOUS transplant is rarely used in AML (no GVL effect).

BOARD TRAP

AML transplant is ALLOGENEIC (need GVL from a donor), not autologous; allo-HCT in CR1 is for intermediate/adverse risk, not favorable.

7. Donor types

WHAT YOU SEE

Three donor portraits tagged MATCHED, HAPLO (parent/child), and CORD — the gallery of donors is the matched-sibling → unrelated → alternative-donor hierarchy.

WHAT IT ENCODES

Donor hierarchy: HLA-matched sibling, then matched unrelated donor, then alternative donors — haploidentical (a half-matched relative, parent/child/sibling, using post-transplant cyclophosphamide) or umbilical cord blood. Alternative donors have expanded transplant access so that lack of a matched donor rarely precludes HCT.

BOARD TRAP

Haploidentical and cord-blood options mean almost every eligible patient can find a donor — 'no matched sibling' is not a reason to deny allo-HCT in adverse-risk AML.

8. Graft-vs-leukemia

WHAT YOU SEE

A knight (GVL) defending the bone-marrow plant from attackers — the defending knight is the donor T-cell graft-versus-leukemia effect that makes allo curative.

WHAT IT ENCODES

The graft-versus-leukemia (GVL) effect — donor T cells recognizing residual leukemic cells — is the immunologic basis for the curative benefit of allo-HCT and the reason autologous transplant is inferior in AML. Donor lymphocyte infusion can boost GVL at relapse.

BOARD TRAP

GVL is the curative engine of allo-HCT; it is inseparable from GVHD risk, and overly aggressive immunosuppression that blunts GVHD can also blunt GVL and raise relapse.

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9. GVHD

WHAT YOU SEE

Tipping scales labeled GVHD with a hostile figure attacking the host — the imbalanced scales are graft-versus-host disease, the flip side of GVL.

WHAT IT ENCODES

Graft-versus-host disease (donor immune cells attacking host tissues — skin, gut, liver) is the principal toxicity of allo-HCT. Acute GVHD targets skin/GI/liver; chronic GVHD resembles autoimmune disease. Prophylaxis: calcineurin inhibitor + methotrexate, or post-transplant cyclophosphamide. GVHD and GVL are two sides of the same alloimmune coin.

BOARD TRAP

More GVHD prophylaxis lowers GVHD but can raise relapse by reducing GVL — the balance is the central trade-off of allo-HCT.

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10. Age <75 eligibility

WHAT YOU SEE

A shield stamped <75 on the tower — the age shield is a soft cutoff: fitness/comorbidity, not chronologic age, decides transplant eligibility.

WHAT IT ENCODES

Transplant eligibility hinges on performance status, comorbidities, and organ function more than chronologic age; reduced-intensity conditioning extends allo-HCT to older/less-fit patients (often into the 70s). It is fitness, not a hard age cutoff, that decides.

BOARD TRAP

Age alone does not disqualify a patient from allo-HCT — reduced-intensity conditioning allows transplant in fit older adults; use comorbidity/performance status, not a rigid age number.

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11. FLT3 TKI maintenance

WHAT YOU SEE

A pill bottle labeled FLT3 TKI in the top MAINTENANCE band — the maintenance bottle is a post-transplant FLT3 inhibitor for FLT3-mutated AML.

WHAT IT ENCODES

In FLT3-mutated AML, a FLT3 inhibitor as post-transplant or post-consolidation maintenance (e.g. sorafenib, or midostaurin/gilteritinib in trials) reduces relapse and improves survival.

BOARD TRAP

FLT3-targeted maintenance is specific to FLT3-mutated AML; it is not given to FLT3-wild-type patients.

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12. Oral azacitidine maintenance

WHAT YOU SEE

An ORAL AZA jar in the MAINTENANCE band tagged POST-INDUCTION — the oral pill jar is CC-486/Onureg maintenance for non-transplant patients in CR.

WHAT IT ENCODES

Oral azacitidine (CC-486, Onureg) maintenance after induction ± consolidation prolongs overall and relapse-free survival in patients in CR who are NOT proceeding to transplant (QUAZAR trial). Note CC-486 (oral) is not bioequivalent to injectable azacitidine.

BOARD TRAP

Oral azacitidine (CC-486/Onureg) is a maintenance drug for non-transplant patients in CR — it is distinct from and not interchangeable with subcutaneous/IV azacitidine used as primary therapy.

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13. CR2 → HCT

WHAT YOU SEE

A note on the outer wall reading AIM CR2 → HCT — the directive is to reinduce relapsed AML to CR2, then proceed to curative allo-transplant.

WHAT IT ENCODES

In relapsed AML, the strategy is to reinduce to a second complete remission (CR2) and then proceed to allogeneic HCT, which offers the only realistic chance of cure in the relapsed setting. Targeted salvage (e.g. gilteritinib for FLT3, IDH inhibitors) can serve as a bridge to transplant.

BOARD TRAP

For relapsed AML, the goal is CR2 followed by allo-HCT — chemotherapy alone after relapse is not curative; the transplant is the curative step.

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14. Drug-sensitivity timing (>12 vs <12 mo)

WHAT YOU SEE

Climbing-vine markers reading >12 MO (drug-sensitive) vs <12 MO (resistance) — the split path is CR1 duration predicting salvage success.

WHAT IT ENCODES

Duration of first remission predicts salvage success: a longer CR1 (relapse >12 months out) implies relative chemosensitivity and a better chance of reattaining CR2, whereas early relapse (<12 months) signals chemoresistance and worse salvage outcomes — favoring clinical trials, targeted agents, and prompt transplant.

BOARD TRAP

Early relapse (<12 months) marks chemoresistant disease — do not expect standard reinduction to work well; move quickly toward targeted salvage and allo-HCT.
